V4 Comprehensive Statistical Summary

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CAR V4 STRATIFIED MODEL AUDIT | idata_car_v4.nc
================================================================
WAIC (Deviance): 4221.50 | LOO-CV: 4316.76
Divergences: 0 | Max R-hat: 1.650

PUBLIC HEALTH INTERPRETATION:
- Smoking: +2.22% risk per 5.3% increase
- Male Interaction: -0.50% additional risk
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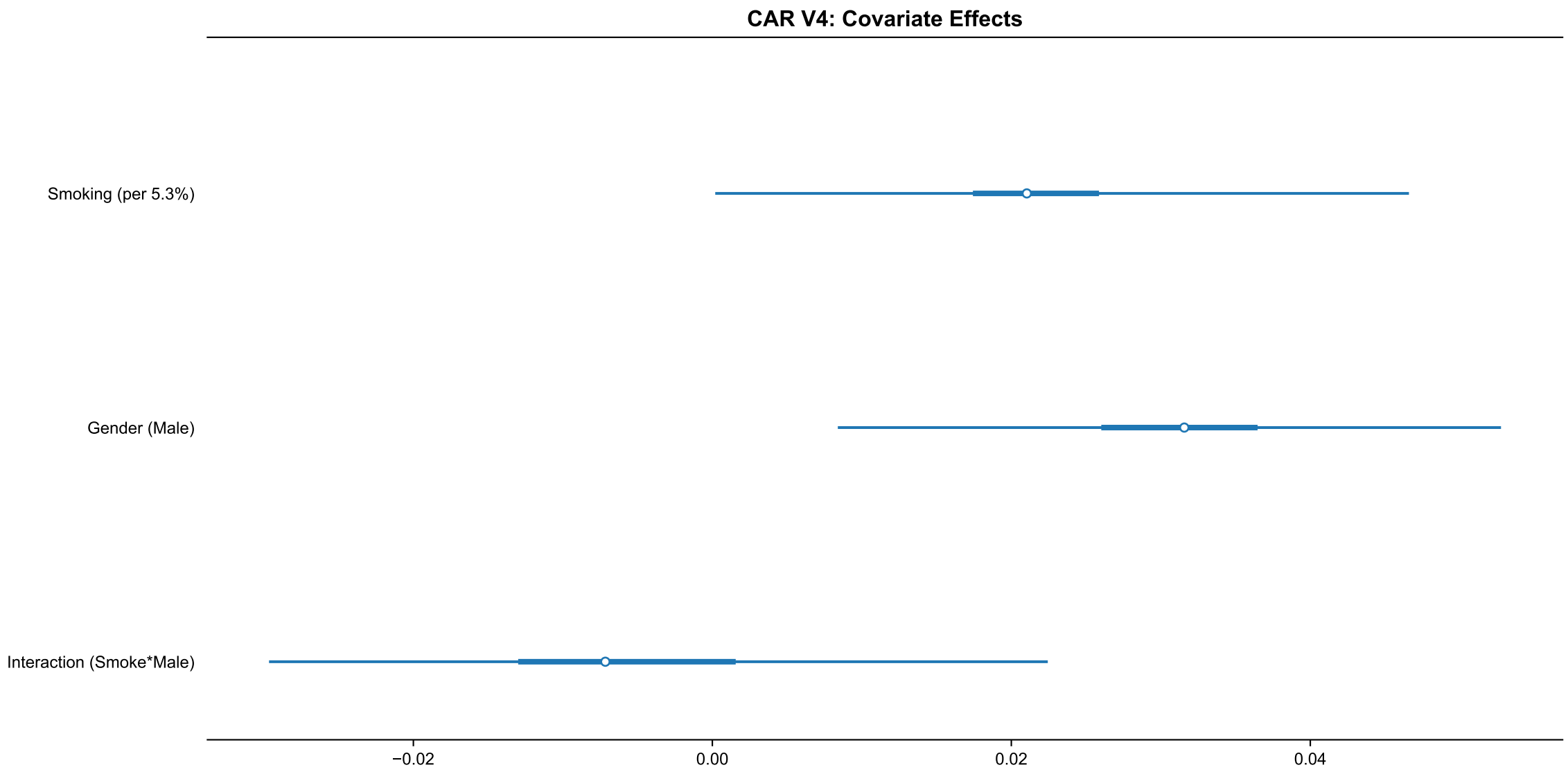
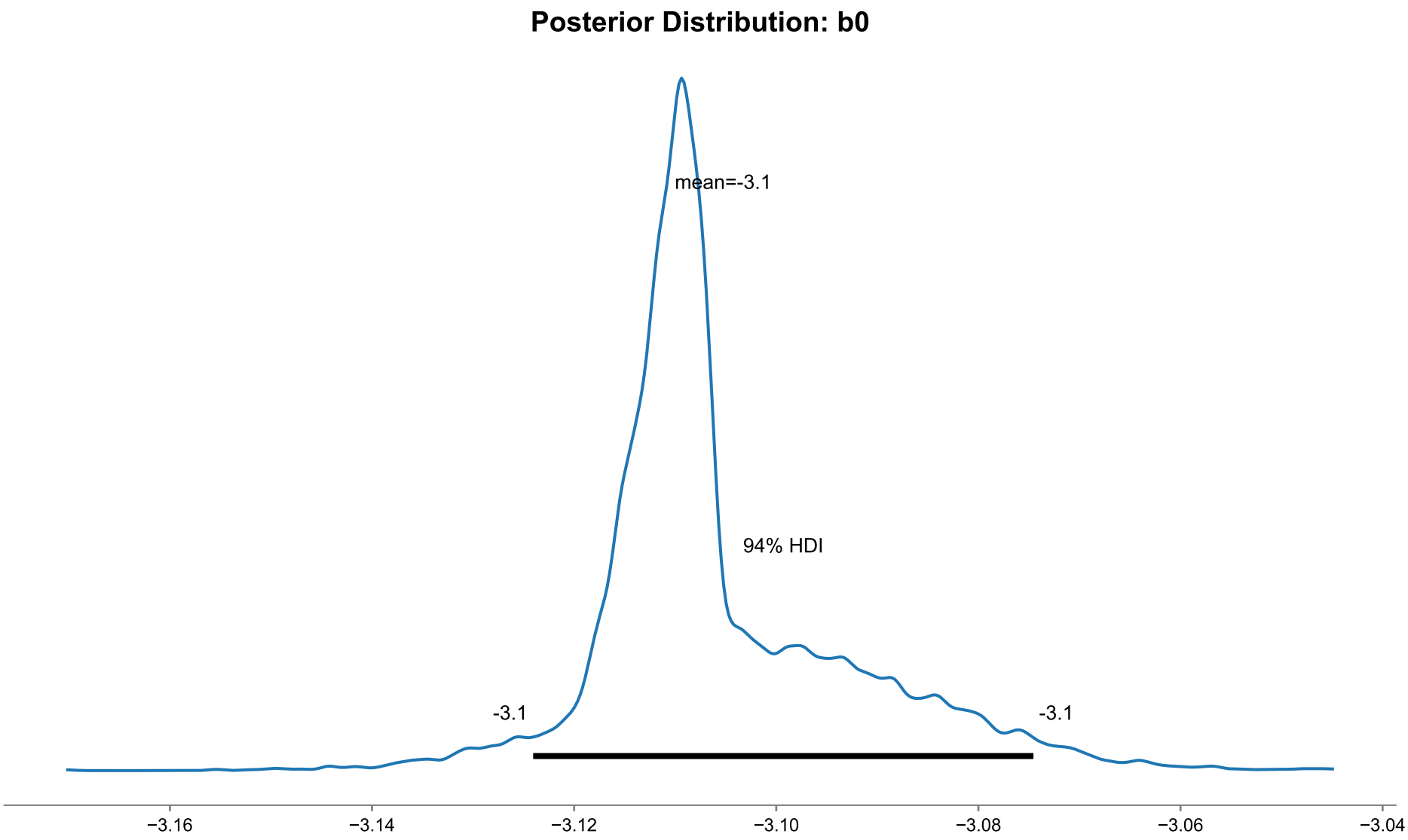
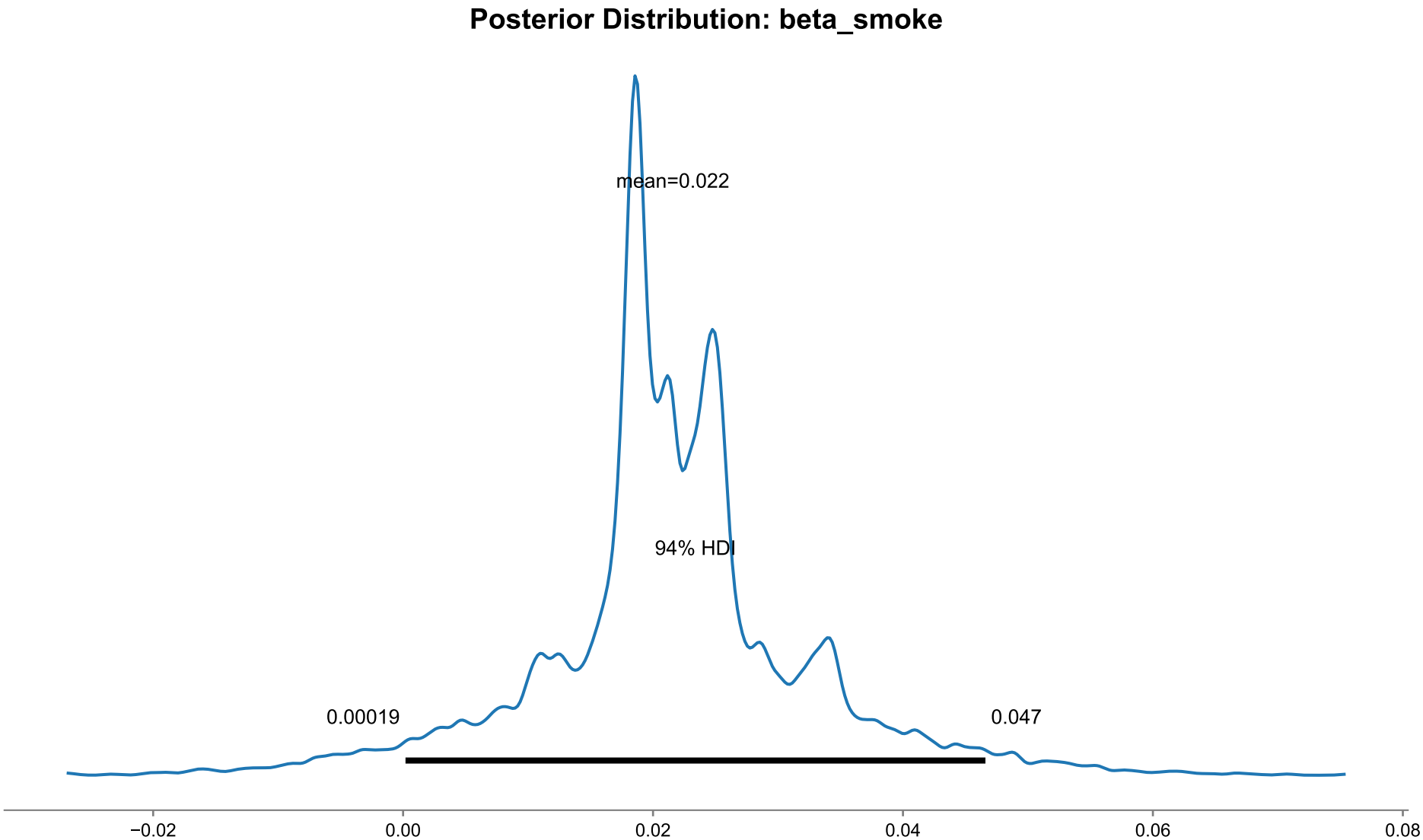
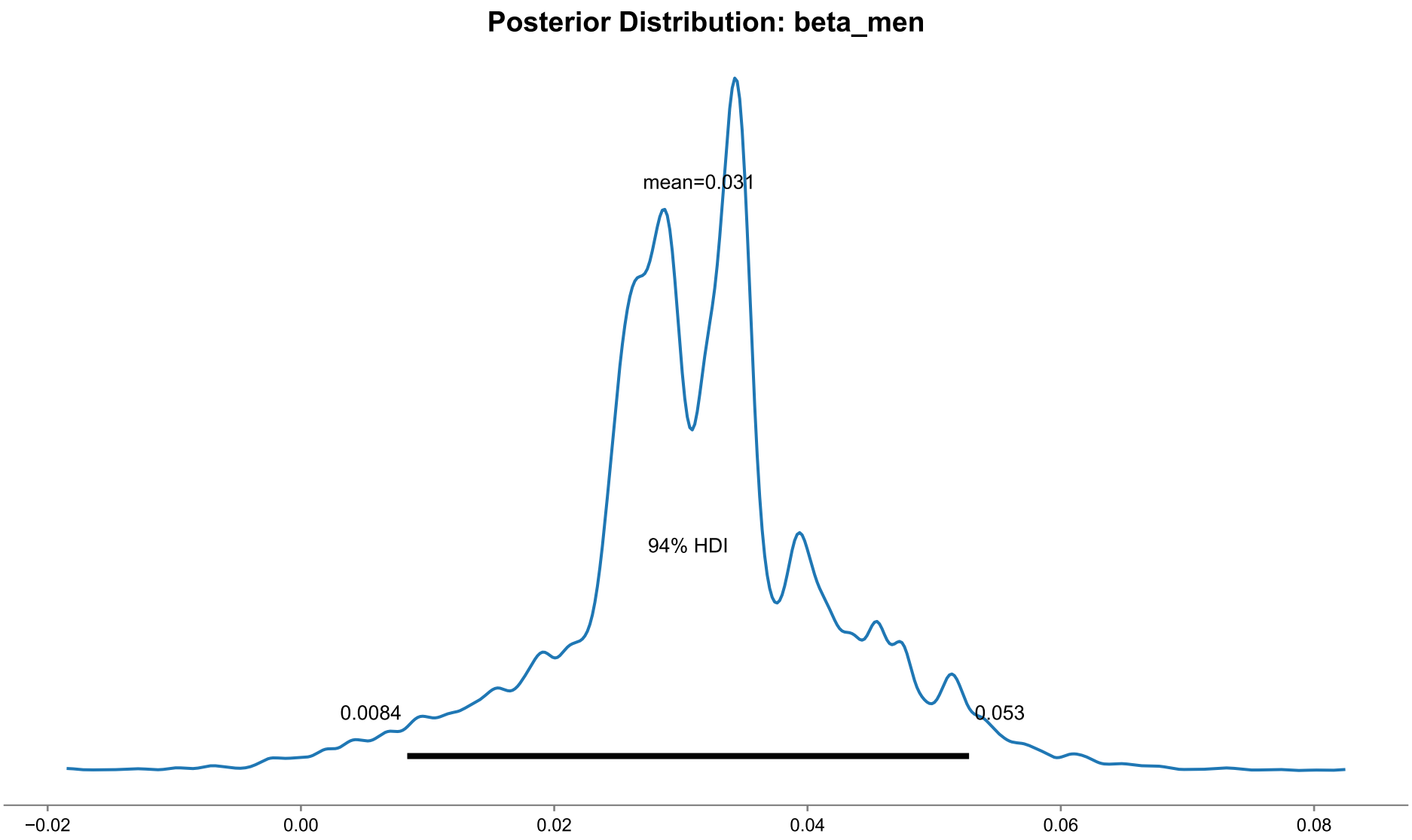
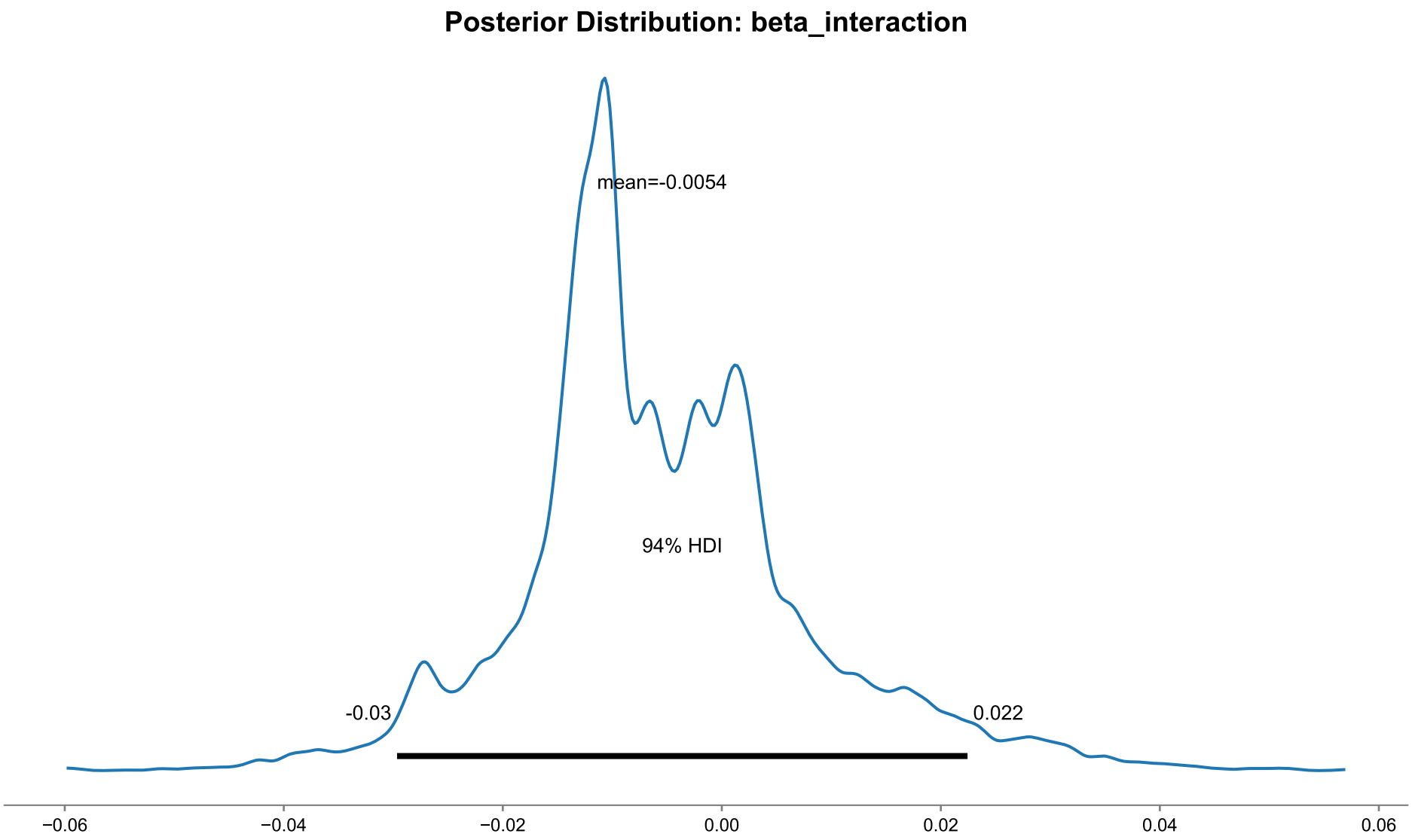
	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
b0	-3.105	0.013	-3.124	-3.075	0.003	0.002	22.0	106.0	1.37
beta_smoke	0.022	0.011	0.000	0.047	0.001	0.002	48.0	95.0	1.19
beta_men	0.031	0.011	0.008	0.053	0.002	0.002	22.0	105.0	1.20
beta_interaction	-0.005	0.013	-0.030	0.022	0.003	0.002	23.0	101.0	1.24
sigma	0.283	0.028	0.230	0.318	0.011	0.003	7.0	54.0	1.65
rho	0.339	0.111	0.077	0.531	0.013	0.025	52.0	147.0	1.61
alpha	0.610	0.233	0.126	0.993	0.056	0.041	21.0	132.0	1.44

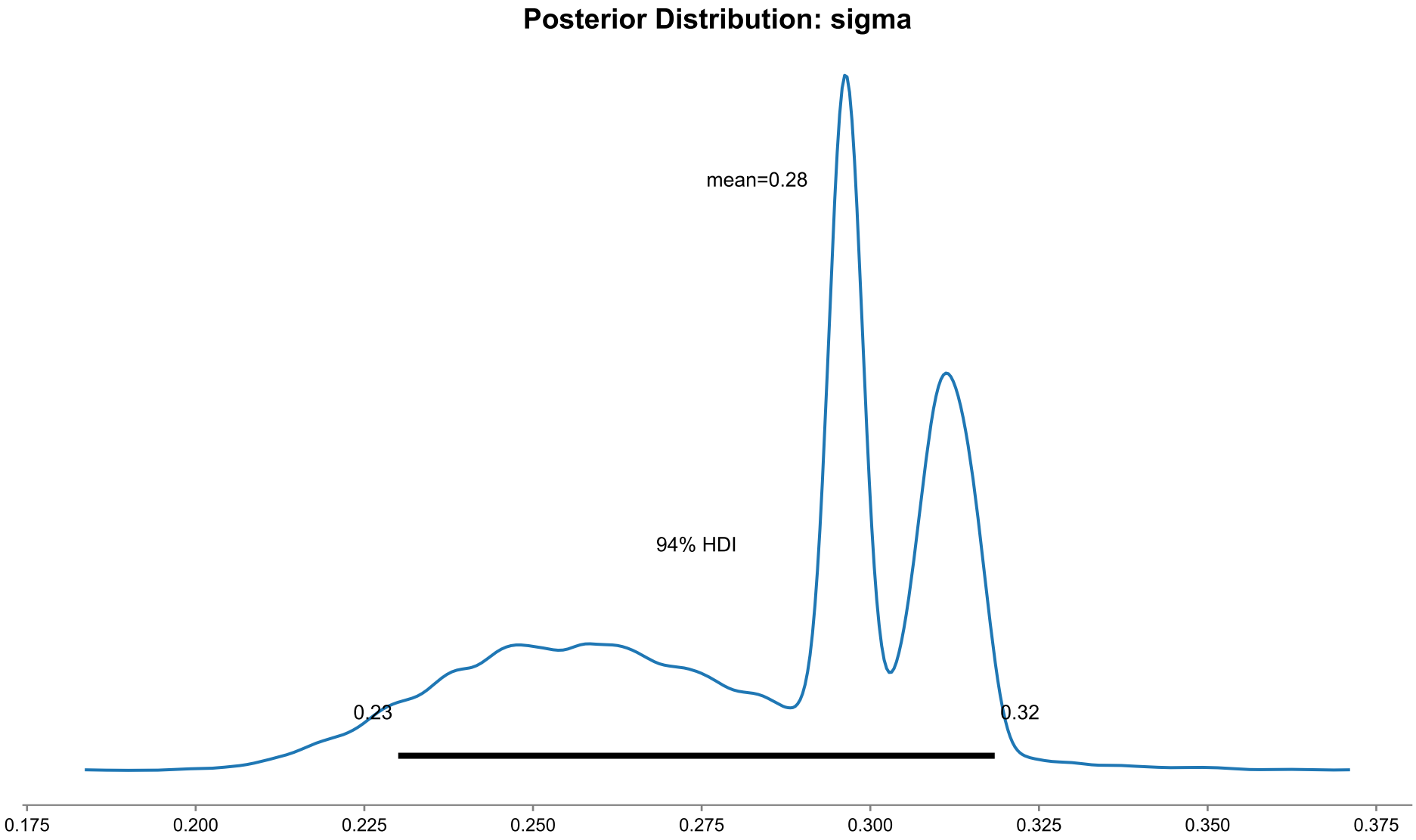

CAR V4: Covariate Effects
Smoking (per 5.3%)
Gender (Male)
Interaction (Smoke*Male)
−0.02
0.00
0.02
0.04

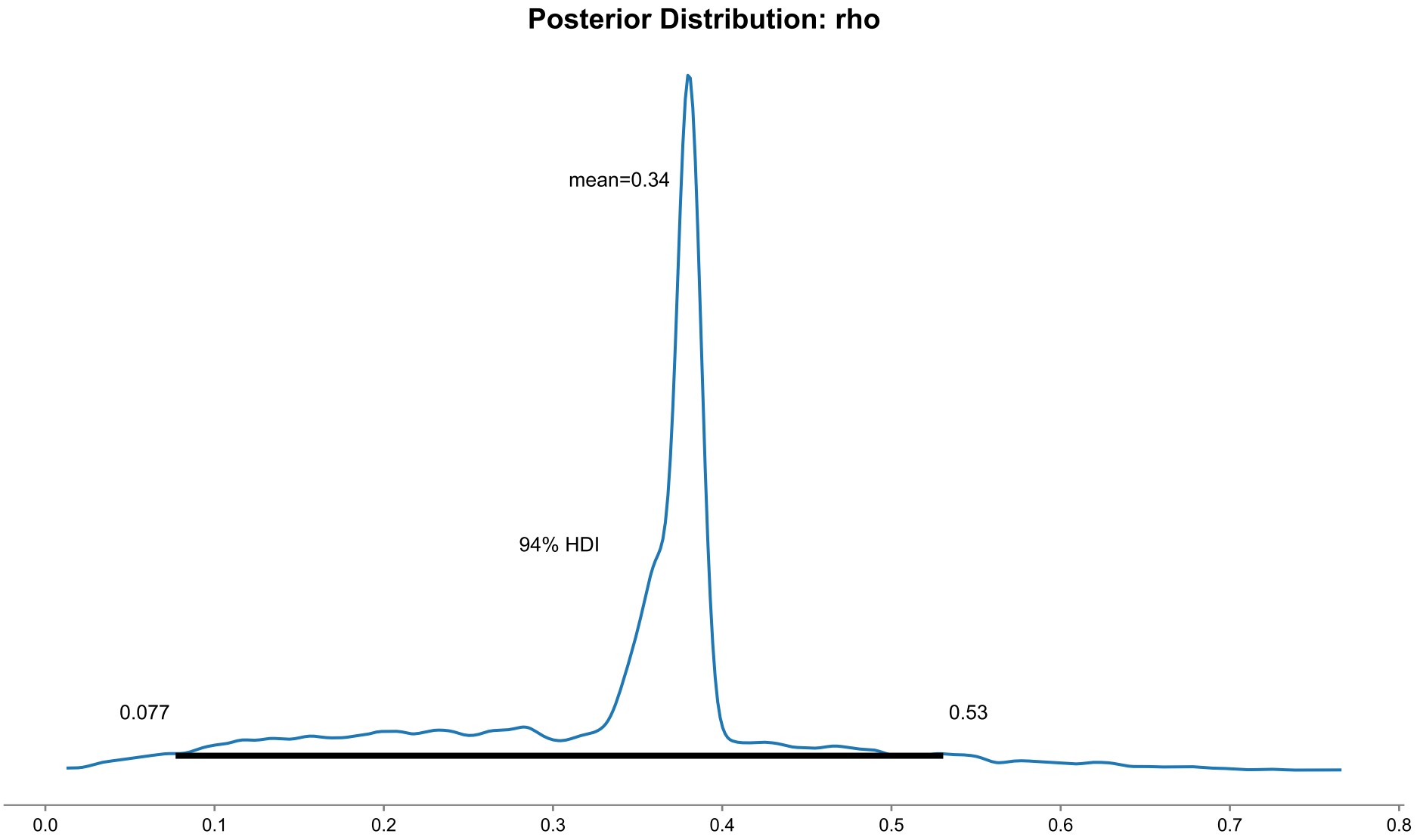

Posterior Distribution: b0
mean=-3.1
94% HDI
-3.1
-3.1
−3.16
−3.14
−3.12
−3.10
−3.08
−3.06
−3.04

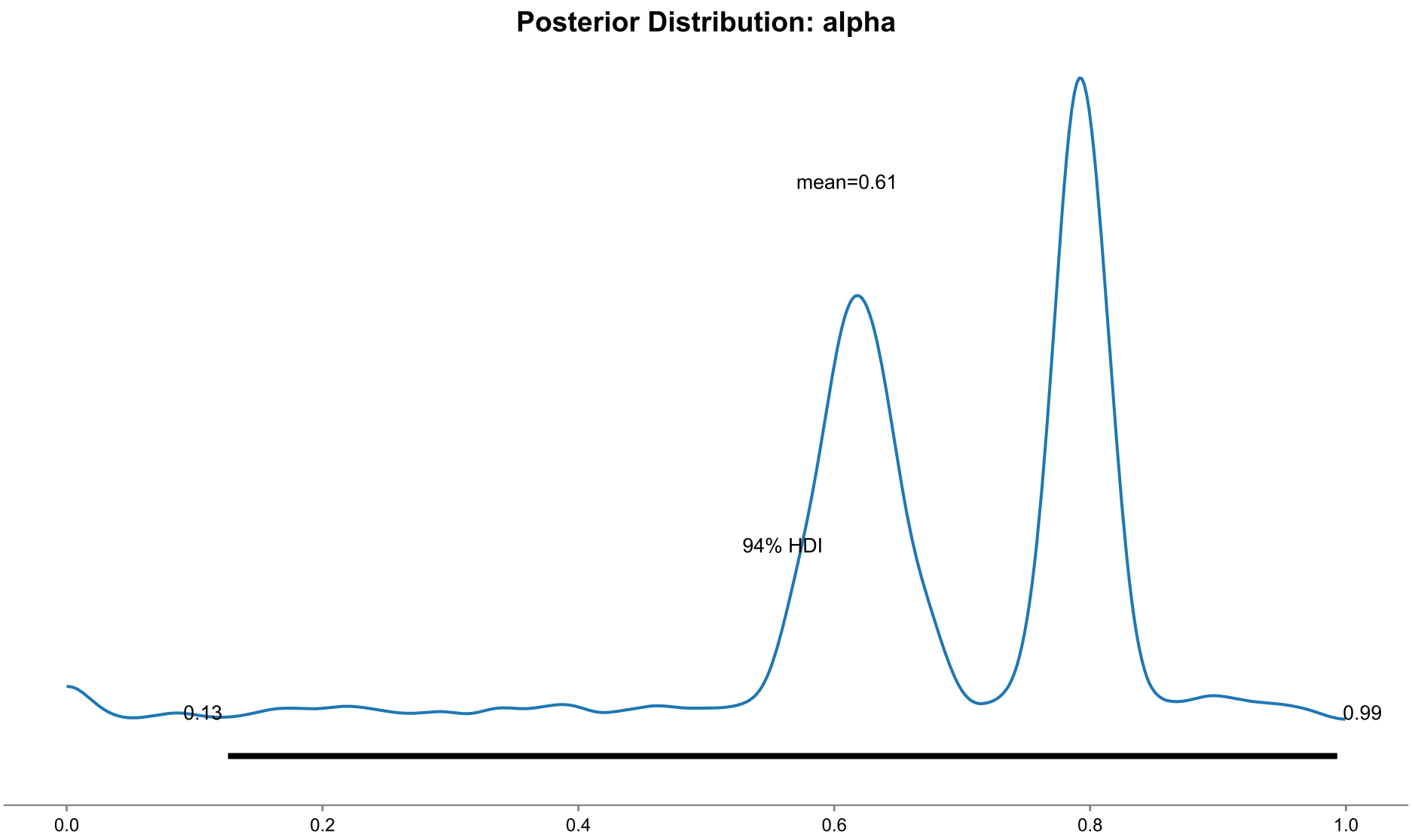

Posterior Distribution: beta_smoke
mean=0.022
94% HDI
0.00019
0.047
−0.02
0.00
0.02
0.04
0.06
0.08

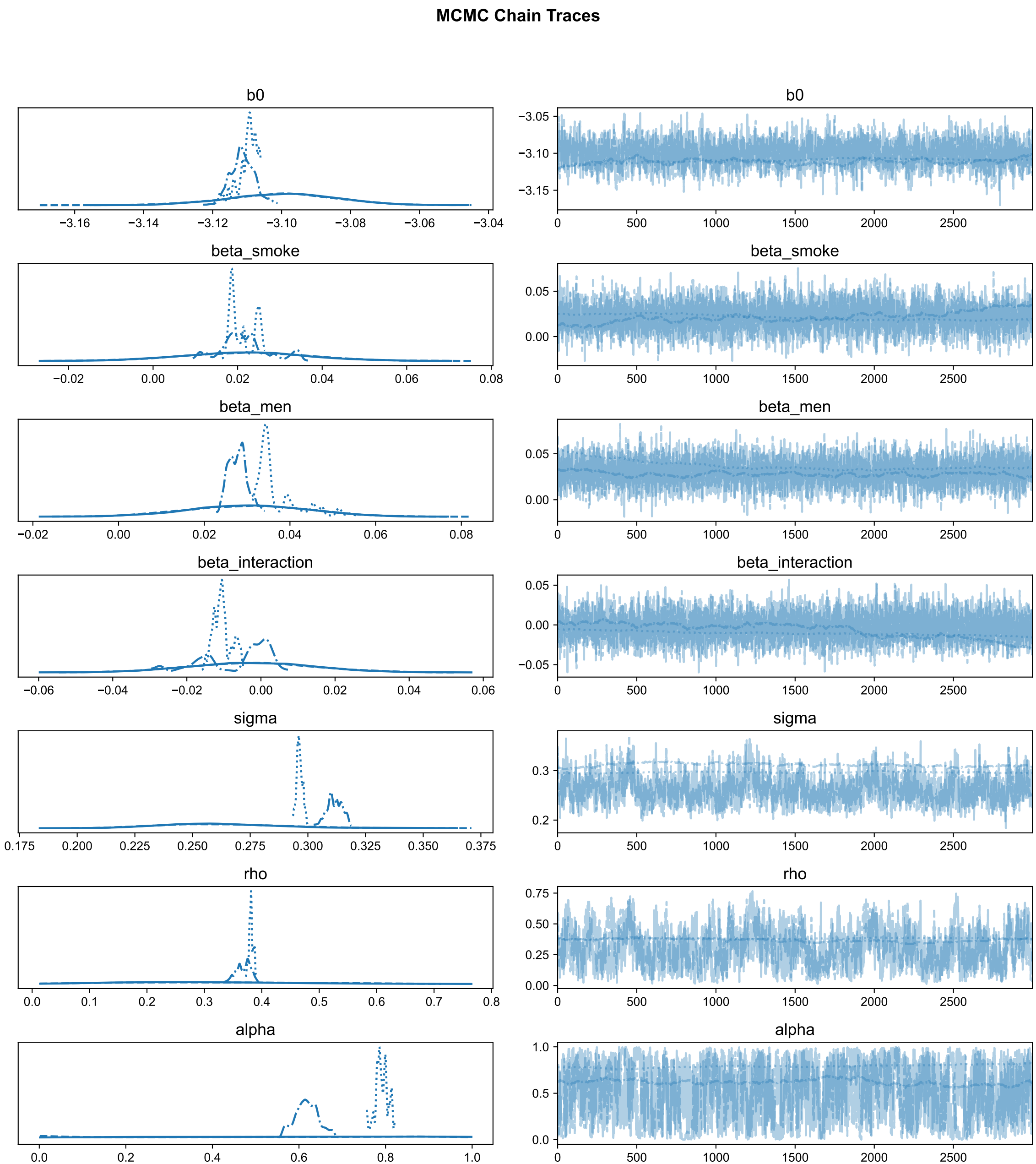

Posterior Distribution: beta_men
mean=0.031
94% HDI
0.0084
0.053
−0.02
0.00
0.02
0.04
0.06
0.08


Posterior Distribution: beta_interaction
mean=-0.0054
94% HDI
-0.03
0.022
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06


Posterior Distribution: sigma
mean=0.28
94% HDI
0.23
0.32
0.175
0.200
0.225
0.250
0.275
0.300
0.325
0.350
0.375


Posterior Distribution: rho
mean=0.34
94% HDI
0.077
0.53
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8


Posterior Distribution: alpha
mean=0.61
94% HDI
0.13
0.99
0.0
0.2
0.4
0.6
0.8
1.0


MCMC Chain Traces
b0
beta_smoke
beta_men
beta_interaction
sigma
rho
alpha